

# **ENV 6128**

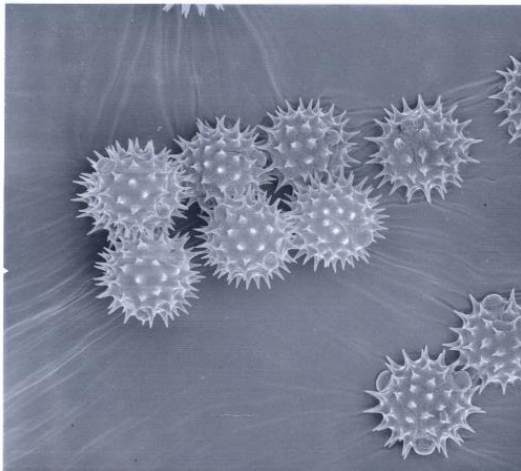
## **Smart Air Quality Monitoring & Air Pollution Control**

# Today's Topics

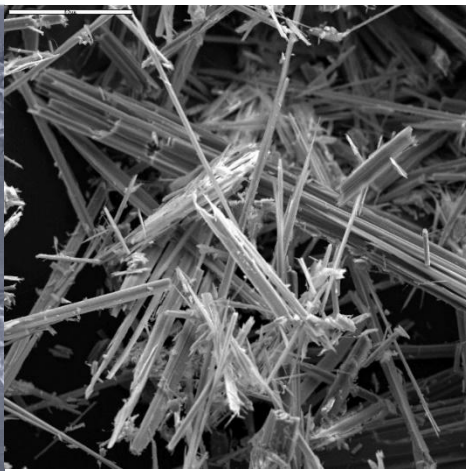
- Intro to atmospheric PM and basic aerosol science
- Expectation
  - Understand size distributions and mode (3-mode model) of atmospheric PM, major sources of atmospheric PMs and their interactions
  - Understand the behavior of aerosol

# Particulate matter

- PM are everywhere in the environment
- Vary greatly in source, size and composition



pollen



asbestos



volcanic ash

# Terminology

- Particulate matter (PM): complex mixture of very small particles and liquid droplets
- Aerosol: ensemble of suspended PMs
- Total suspended particles (TSP) - all suspended PMs
- Coarse particles: PMs with diameter  $> 2.5 \mu\text{m}$
- Inhalable coarse particles:  $\text{PM}_{10}$ - $\text{PM}_{2.5}$
- Fine particles:  $\text{PM}_{2.5}$
- Ultrafine particles:  $\text{PM}_{0.1}$

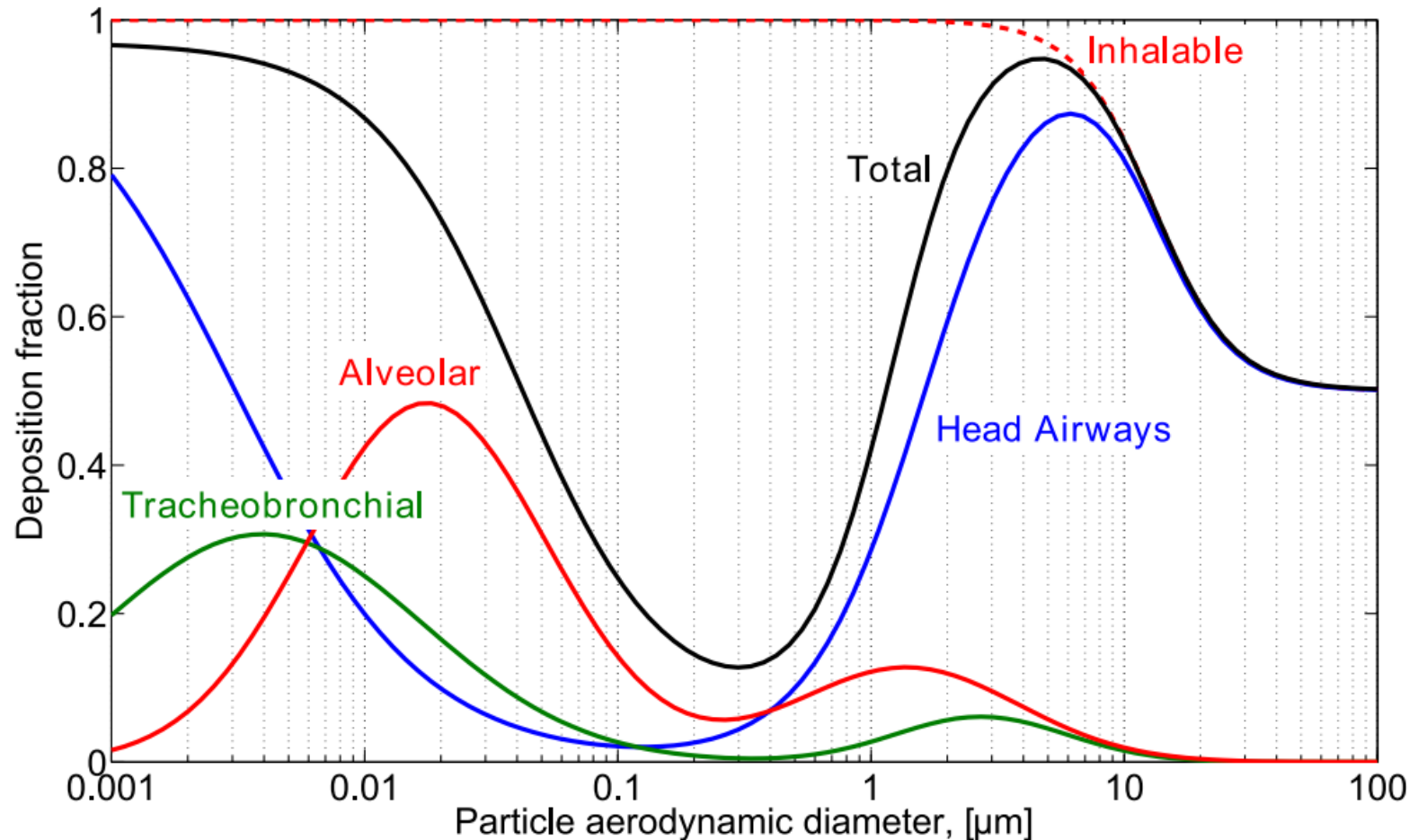
# Concentration Metrics

- Three metrics depending on purpose
  - Number concentration
    - e.g. light scattering and absorption
  - Area concentration
    - e.g. surface reactions
    - Area of a spherical particle:  $a_p = 4\pi r^2 = \pi D_p^2$
  - Mass concentration
    - e.g. health effect due to mass dose
    - Volume of a spherical particle:  $v_p = 4/3\pi r^3 = 1/6\pi D_p^3$

# Particulate matter

- For aerosol, size matters
  - Different size particle behavior differently
  - Smaller size particles generally:
    - Suspend in the air longer
    - Goes deeper inside lungs (sort of)
- Size for non-spherical aerosol
  - Most used: aerodynamic diameter ( $D_a$ )
  - The diameter of a round water droplet ( $1\text{g/cm}^3$ ) that has the same settling velocity as the particle

# Particle deposition in human respiratory tract



# ACGIH sampling criteria

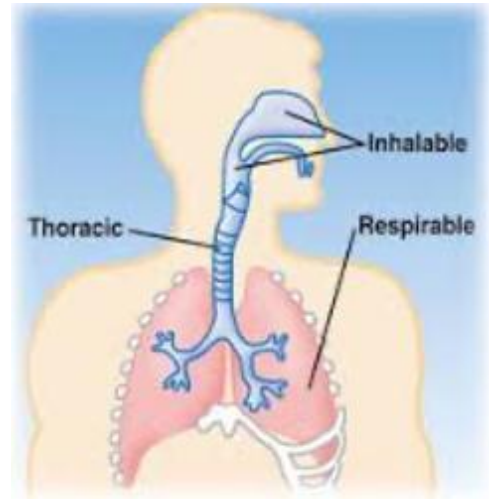
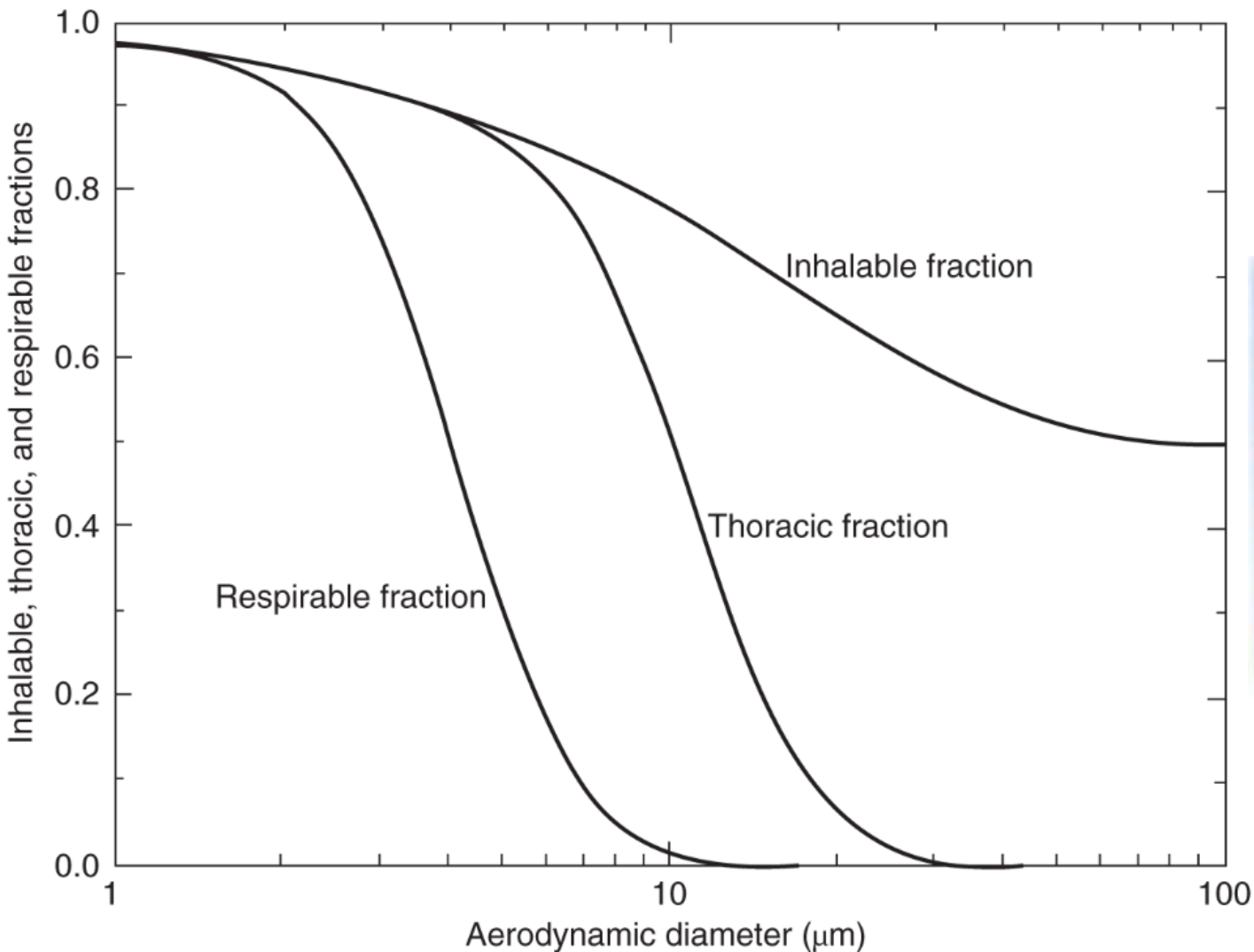


Figure 11.5. Hinds & Zhang. 2023. ACGIH (American Conference of Governmental Industrial Hygienists) sampling criteria for inhalable, thoracic, and respirable fractions.

# Particle inhalation

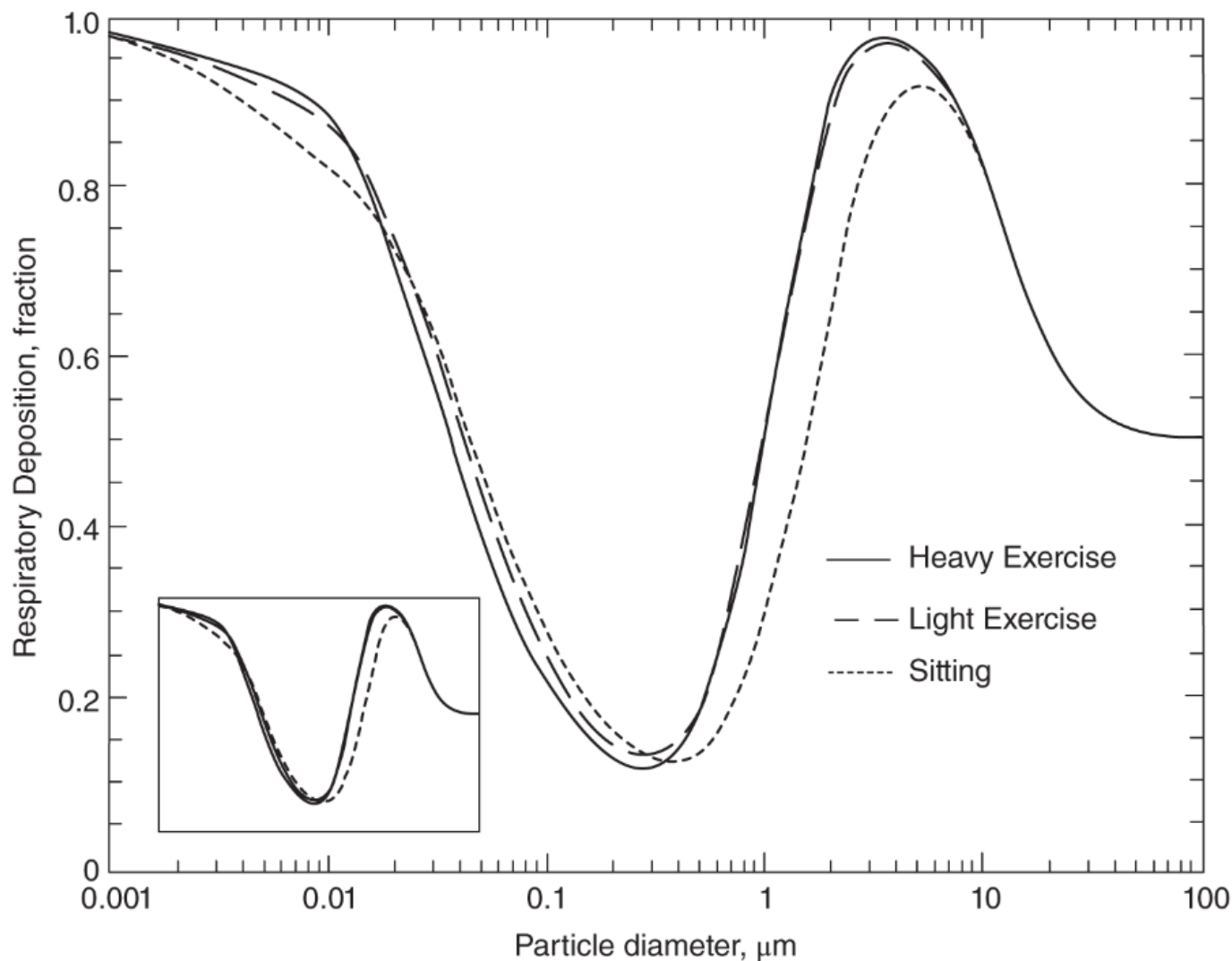


Figure 11.2. Hinds & Zhang, 2023. Predicted total respiratory deposition at three levels of exercise based on ICRP deposition model. Average data for males and females. Inset does not include the effect of inhalability.

# Exposure route

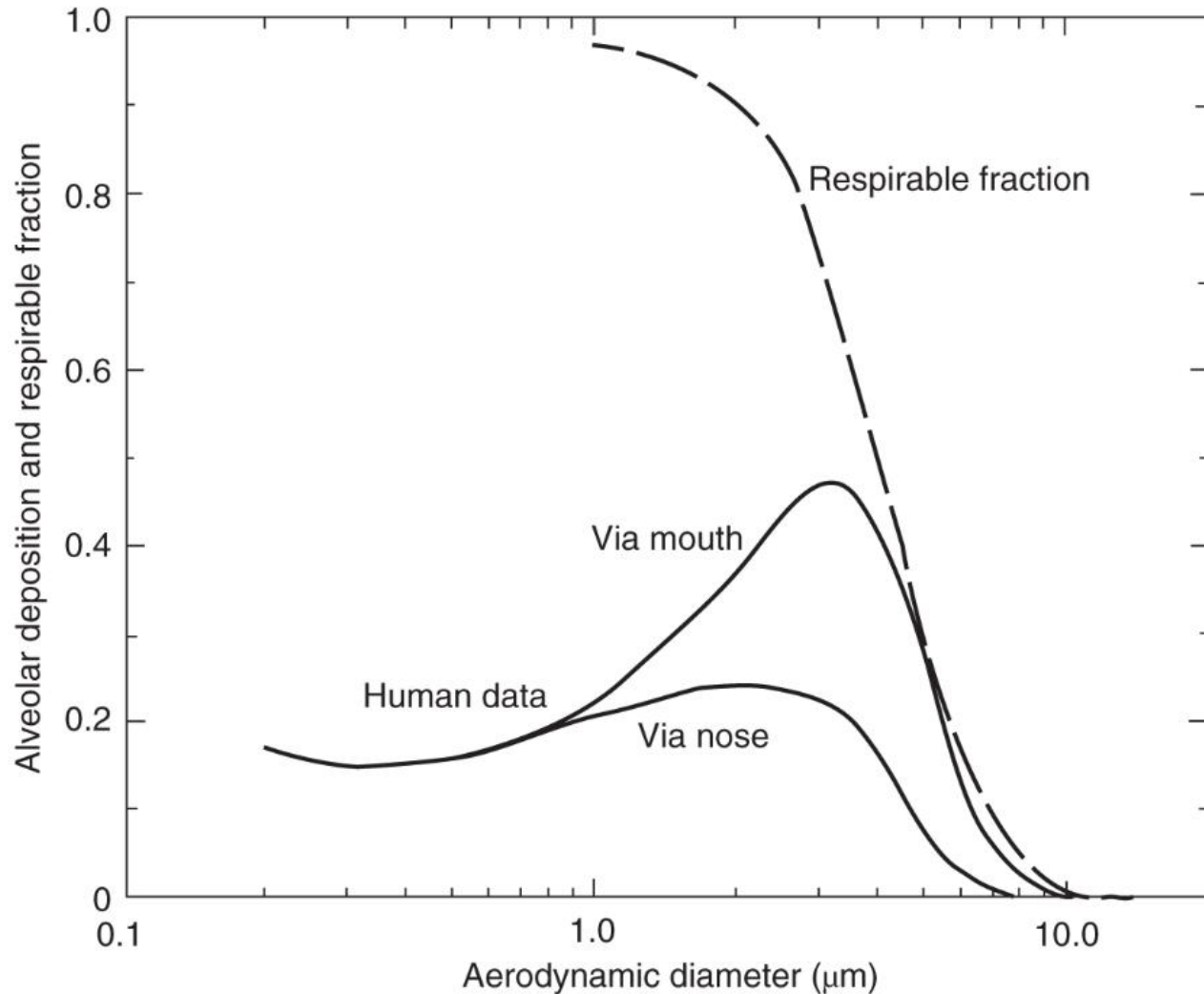
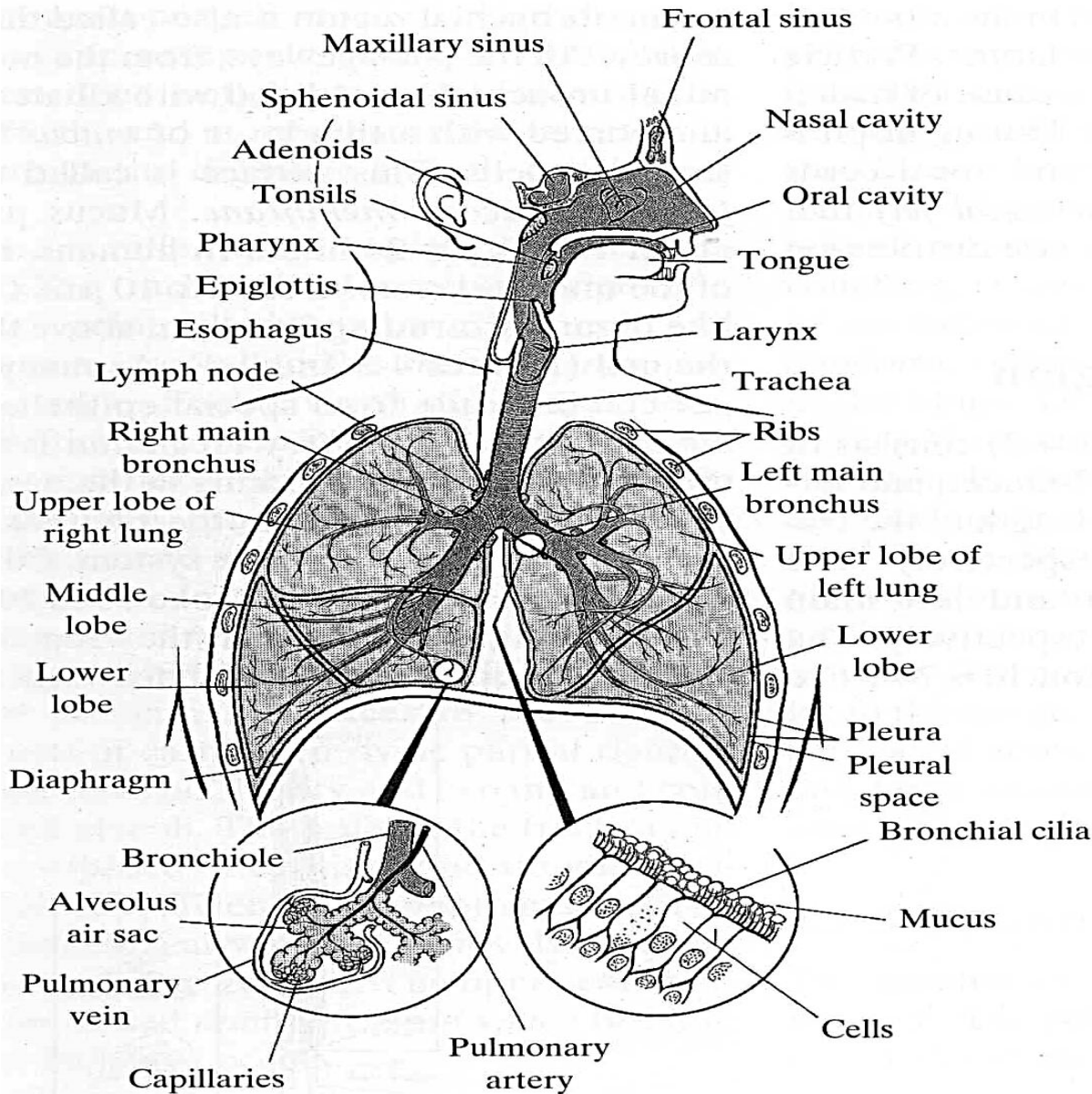


Figure 11.8. Hinds & Zhang. 2023. Comparison of experimental measurements of alveolar deposition (median values) and ACGIH respirable fraction criterion. Human data from Lippmann (1977).

# The respiratory system



*Sources and Control of Air Pollution*, Heinsohn and Kabel, Prentice-Hall, 1999

# The respiratory system

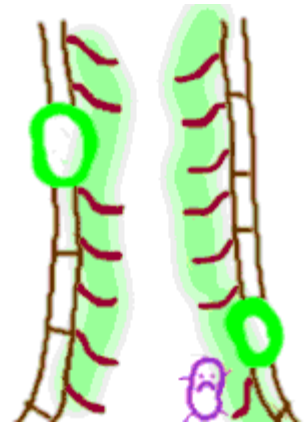
- Consist of upper airway, tracheobronchial tree and pulmonary region
- Overall function
  - Gas to blood exchange
  - Supply the body with  $O_2$  for metabolism
  - Remove waste  $CO_2$
  - Removal of other volatile toxicants
  - Physical immune response
    - Eliminating of viruses, etc before they get to the blood

# The respiratory system

- Upper airway
  - Nose to vocal chords
  - Warm and humidifies air
  - Filtering of particles by hairs
  - Aerodynamic impaction of particles (winding passages)
  - Adsorption in mucus (which is swallowed or spit out)
  - Sneeze reflex
  - Olfaction (smell)

# The respiratory system

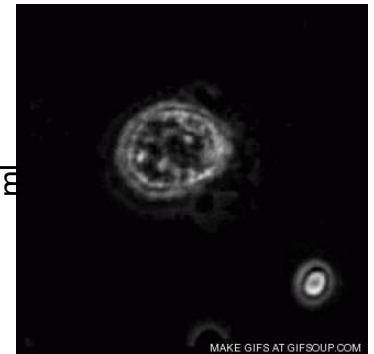
- Tracheobronchial tree
  - Air transport
  - Bronchoconstriction – capture foreign substances
  - Mucociliary escalator – beating cilia move mucus and foreign substances toward mouth/nose
  - Cough reflex



# The respiratory system

- Pulmonary region
  - Respiratory bronchioles, alveolar ducts, alveoli (air sacs)
  - Site of air to blood gas exchange (mass transfer by molecular diffusion)
    - 50-100 m<sup>2</sup> of gas exchange surface
  - Phagocytosis - ingestion of deposited matter by phagocytes and macrophages (from white blood cells)
  - Absorption to lymphatic system

<http://gifsoup.com/view/2305747/phagocytosis.html>



# PM in the respiratory system

- Particles must be deposited to have an effect
- Major factor for particulate behavior is their aerodynamic properties
- Depend on size (and orientation), shape, density
  - Not all particles can be inhaled
  - Some inhaled particles are exhaled

# PM in the respiratory system

- Upper airways:
  - Capture larger particles ( $D > 5\text{-}10\ \mu\text{m}$ ); clear in  $< 1\ \text{d}$
  - Capture very small particles ( $D < 0.01\ \mu\text{m}$ )
  - Mouth breathing allows larger PM to penetrate deeper in lungs
- Pulmonary region
  - Deposition of small particles ( $0.01 < D < 4\ \mu\text{m}$ )
  - Clearance takes weeks to months
  - Particles  $< 0.1\ \mu\text{m}$  are readily absorbed by tissues

# Mode of PM

- Essentially size bins
- Here we use three modes:
  - Nucleation mode ( $D < 0.1\mu\text{m}$ )
  - Accumulation mode ( $2.5\mu\text{m} > D > 0.1\mu\text{m}$ )
  - Coarse mode ( $D > 2.5\mu\text{m}$ )
- Bin # and size ranges definition vary

# Mode of PM

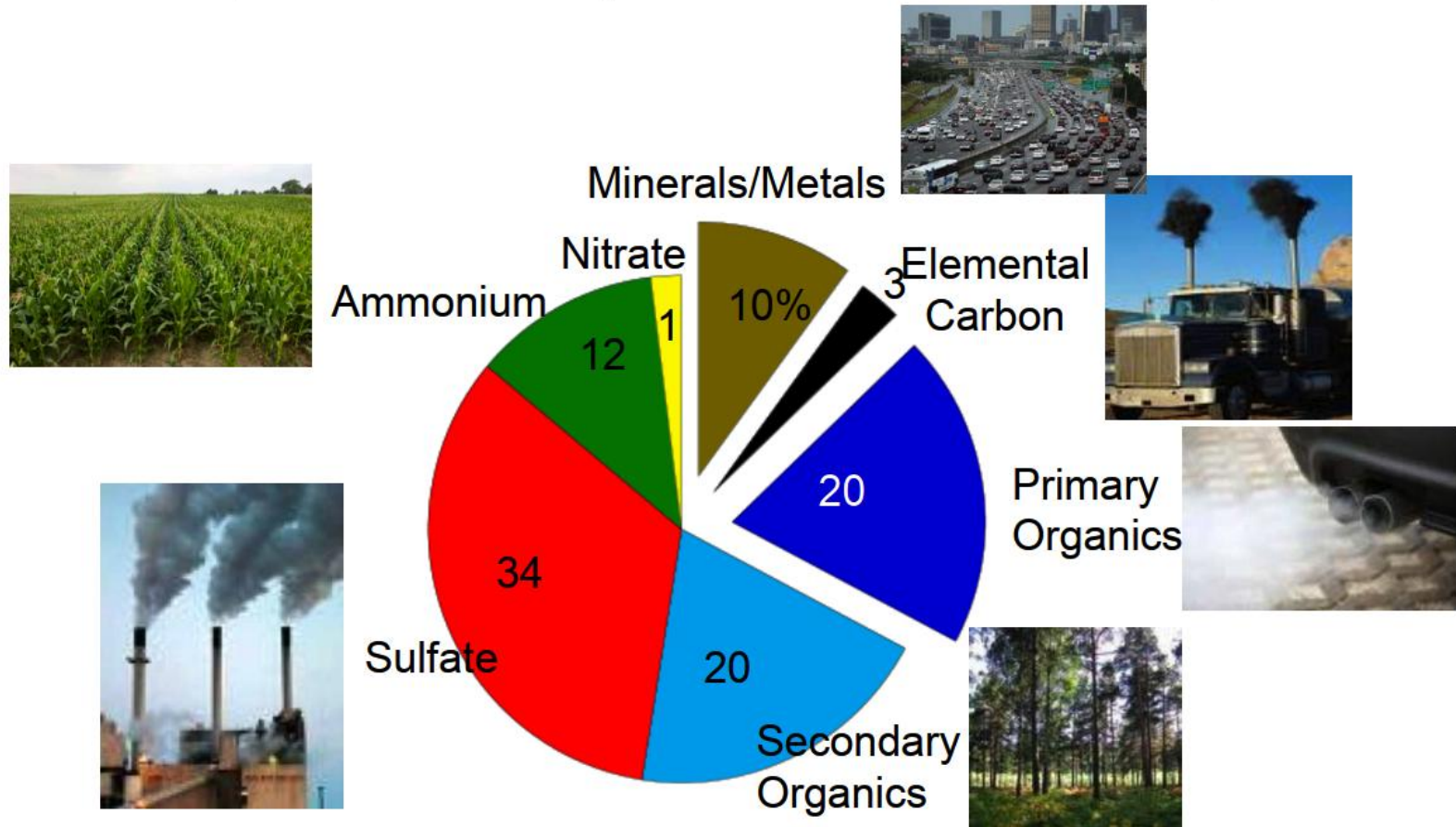
- Nucleation mode ( $D < 0.1\mu\text{m}$ )
  - Mainly combustion PMs, and particles formed in the atmosphere by gas-to-particle conversion.
  - Quickly grow into accumulation mode.
  - Largest number concentration
- Accumulation mode ( $2.5\mu\text{m} > D > 0.1\mu\text{m}$ )
  - Still small but growth can be slow
  - Large surface area concentration

# Mode of PM

- Coarse mode ( $D > 2.5\mu\text{m}$ )
  - Mainly primary emissions
  - Natural emissions mainly coarse mode
  - Also comes from mechanical process
    - Agriculture, surface mining etc
  - Generally sink down within a few days
  - Largest mass concentration
- $D > 10\mu\text{m}$  generally not a health concern

Chemical composition of aerosol vary

Fine Particles (PM<sub>2.5</sub>): Chemical composition depends on sources, ....



# Chemical composition of aerosol vary

Mass concentration and mass fraction of nonrefractory inorganic species and organic components in sub-micrometer aerosols

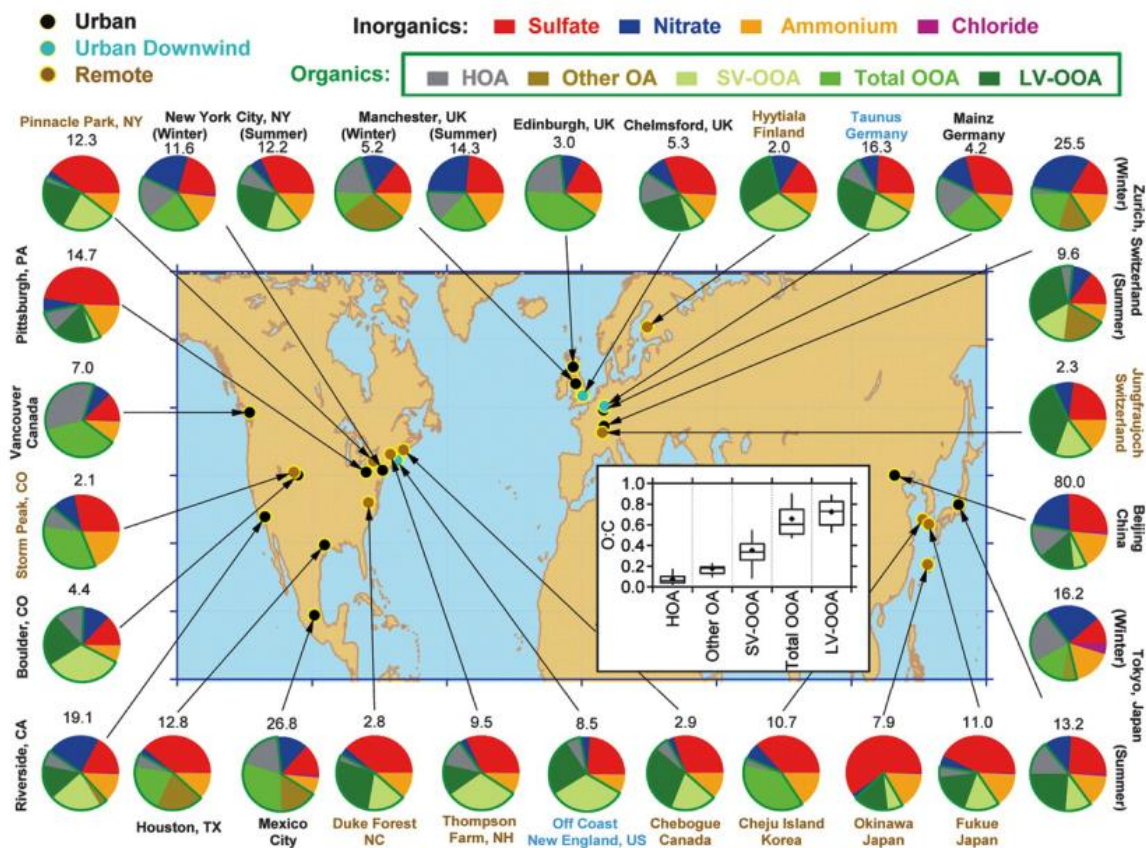


Fig. 1 Total mass concentration (in micrograms per cubic meter) and mass fractions of nonrefractory inorganic species and organic components in submicrometer aerosols measured with the AMS at multiple surface locations in the Northern Hemisphere (21). [Jimenez et al., Science, 2009]

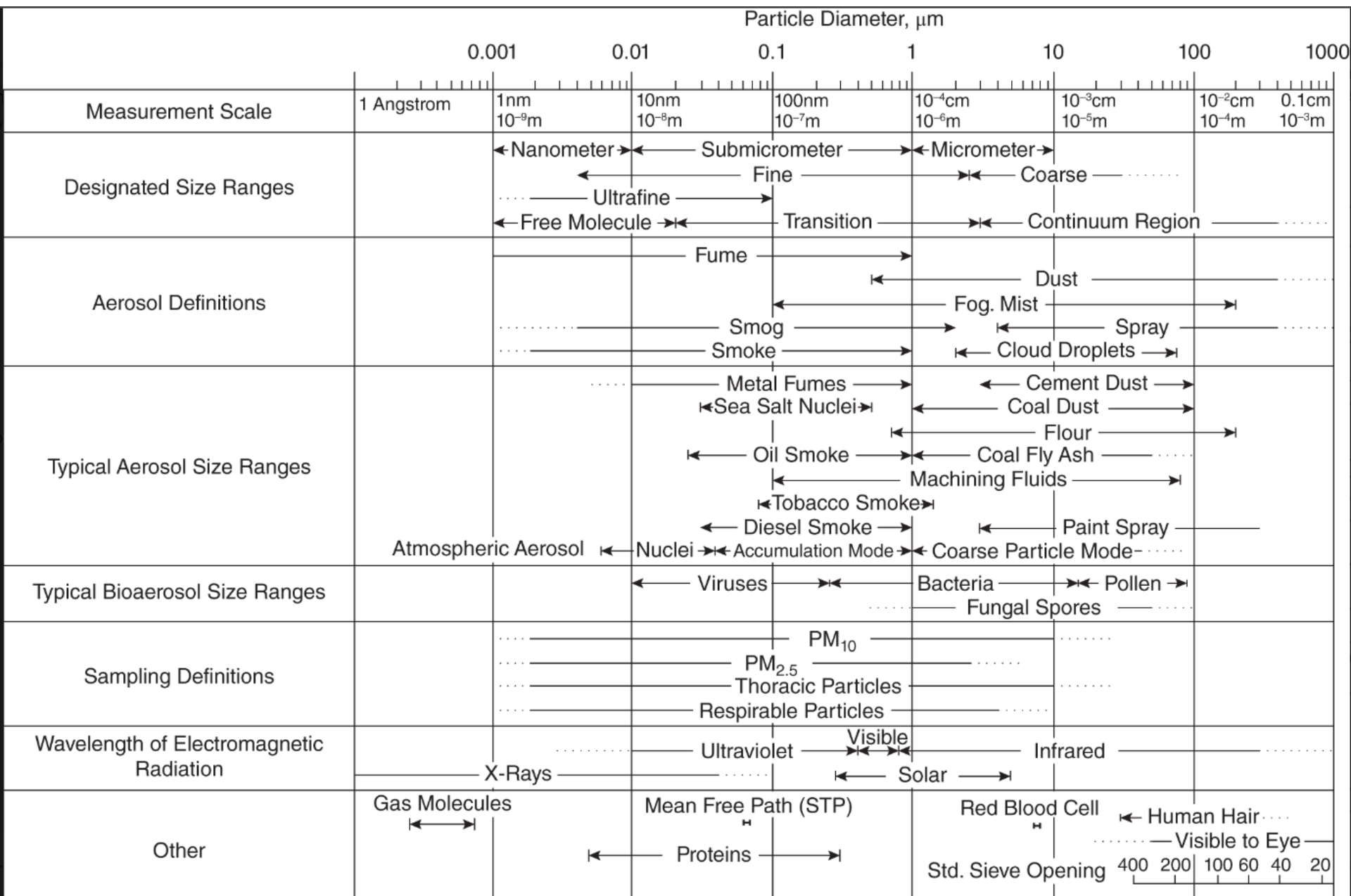
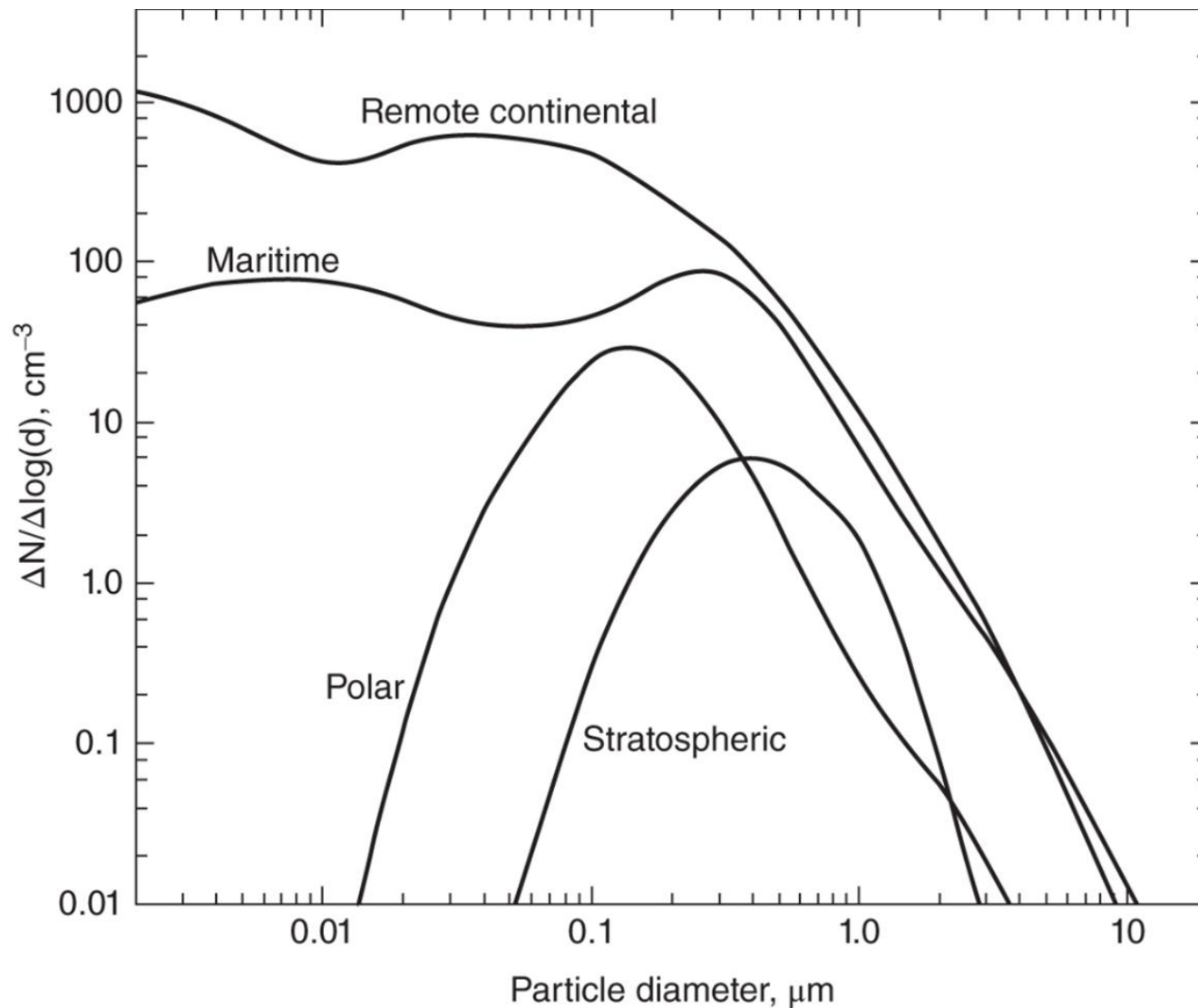


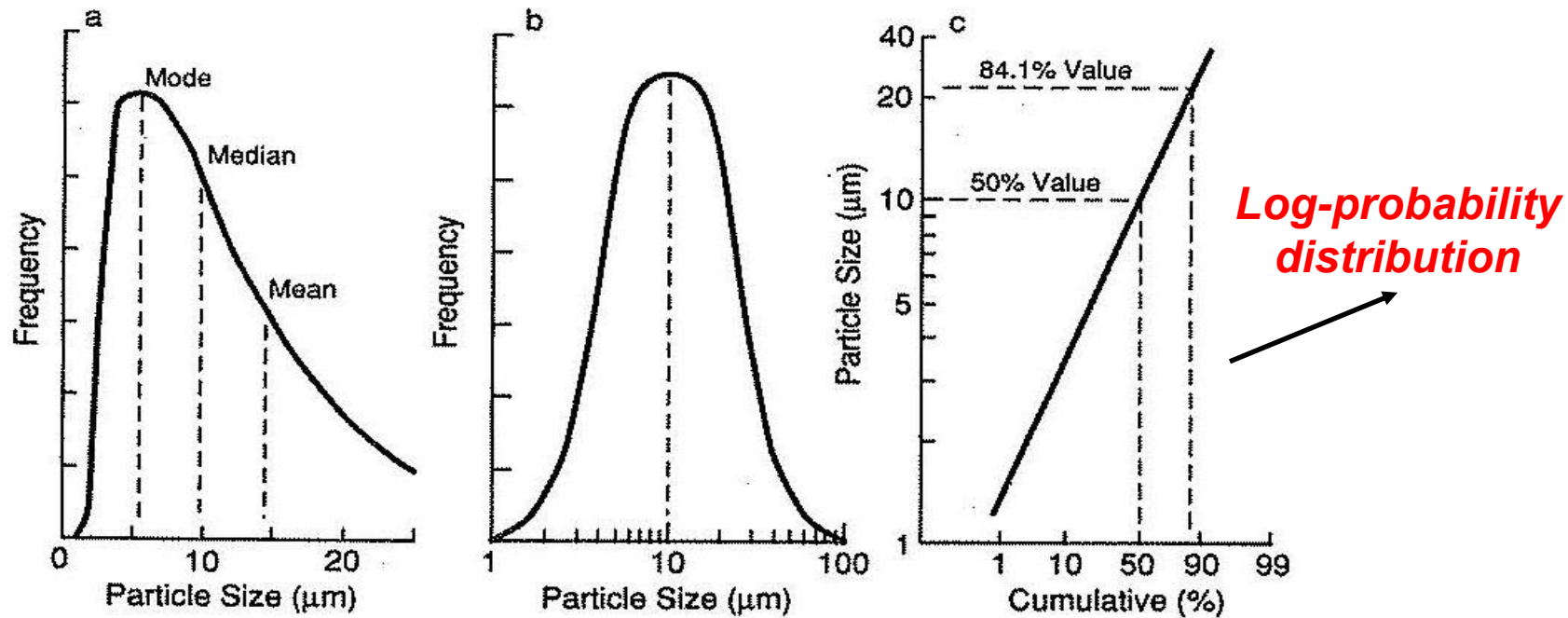
FIGURE 1.6. Hinds &amp; Zhang. 2023. Particle size ranges and definitions for aerosols.

# Background atmospheric aerosols size



# Size distribution of aerosols

- A sample of aerosols usually contains particles of different sizes

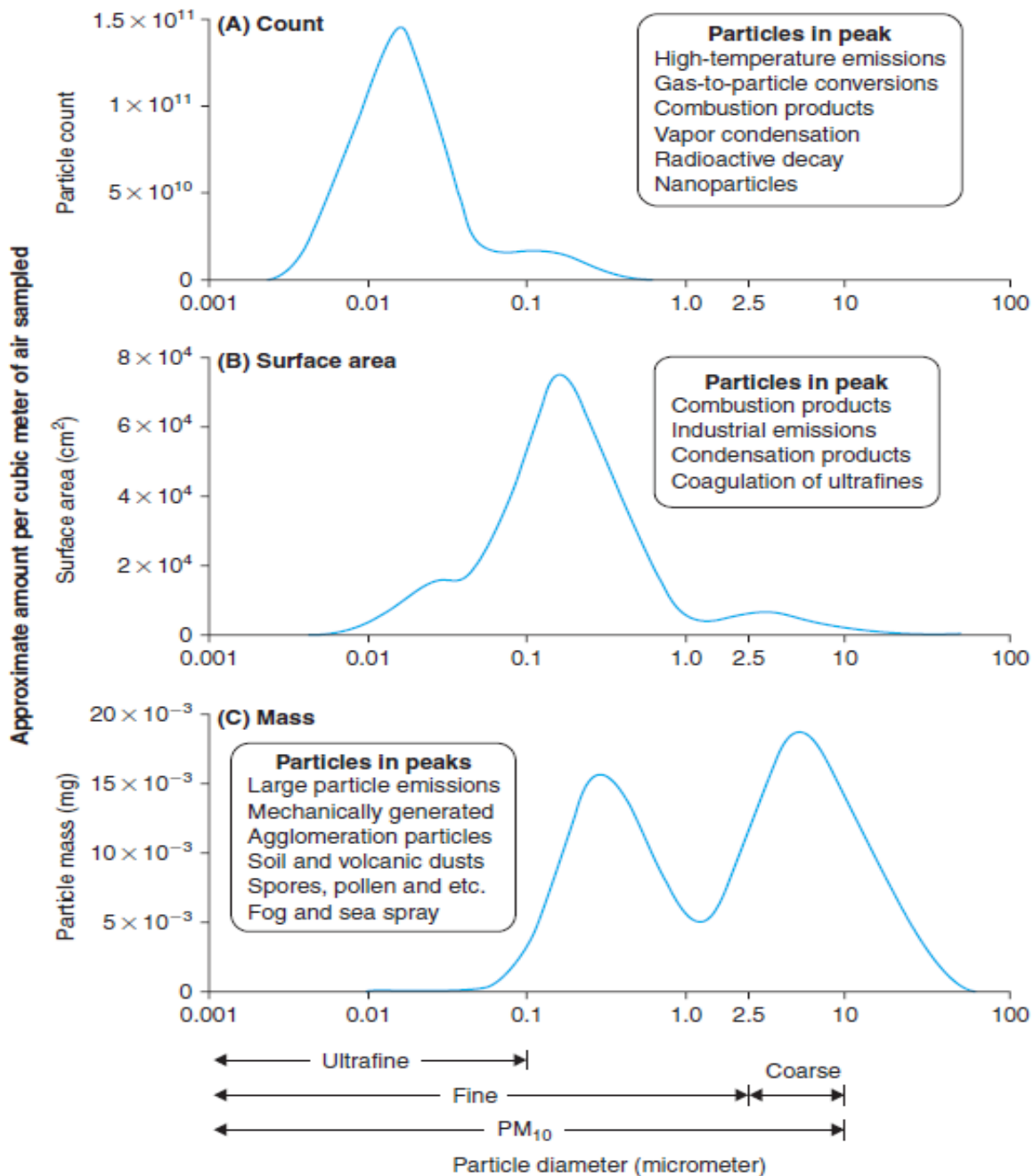


Differential and cumulative size distributions.

Source: Lippmann, M. (2000) 'Introduction and background'. In: M. Lippmann (ed.) *Environmental Toxicants*, Wiley, New York.

(from Tiwary)

# Example count, surface area and mass distributions of different size aerosols



(Phalen, 2013)

# Source & compositions

**Table 14.1** Sources and Estimates of Global Emissions of Atmospheric Aerosols [a\)](#), [b\)](#).

|                            | Natural Source |         | Anthropogenic Source |         | Total         |         |
|----------------------------|----------------|---------|----------------------|---------|---------------|---------|
| Particle Type              | Range          | Average | Range                | Average | Range         | Average |
| Biogenic primary organic   | 15–70          | 50      | —                    | —       | 15–70         | 50      |
| Biogenic secondary organic | 3–83           | 11      | —                    | —       | 3–83          | 11      |
| Biogenic VOCs              | 835–1000       | 900     | —                    | —       | 835–1000      | 900     |
| Biomass burning organic    | 26–70          | 44      | 12–270               | 90      | 38–340        | 134     |
| Black carbon (soot)        | —              | —       | 8–14                 | 12      | 8–14          | 12      |
| Carbonaceous               | —              | —       | 15–90                | 29      | 15–90         | 29      |
| Industrial dust            | —              | —       | 40–130               | 100     | 40–130        | 100     |
| Mineral dust               | 1000–2150      | 1731    | 50–250               | 150     | 1050–2400     | 1881    |
| Nitrates                   | 12–27          | 25      | 90–118               | 95      | 102–145       | 120     |
| Sea salt                   | 3000–20,000    | 7068    | —                    | —       | 3000 – 20,000 | 7068    |
| Stratospheric volcanic     | 2–17           | 10      | —                    | —       | 2–17          | 10      |
| Sulfates                   | 107–374        | 165     | 50–122               | 73      | 157–496       | 238     |
| Tropospheric volcanic      | 4–90           | 30      | —                    | —       | 4–90          | 30      |
| Total Emissions            | 5004–23,881    | 10034   | 265–994              | 408     | 5269–24,875   | 10442   |

[a\)](#) Adapted from Tomasi et al. (2016).

[b\)](#) Units are in Tg/yr [ $10^6$  metric tons/yr].

# Sea spray

- Largest natural source aerosol source
- Mainly coarse
- Wind → bubbles → spray
- Main composition
  - Seawater with some sodium chloride (NaCl)



# Soil dust

- Natural soil dust
  - Main compositions: mineral and organic materials
  - Mainly coarse
  - Desert and damaged land cover area
- Anthropogenic soil dust
  - Agriculture
  - Re-suspended dust
  - Construction, demolition etc.



# Volcanic dust

- Mainly coarse silicate materials
- Secondary PMs can form from volcanic gases

Mount Saint Helen, 1980



# Biomass burning

- Biomass burning PMs containing plant debris, ash, soil dust, soot etc., and are mainly coarse
- Also emit gases such as phenol, cause secondary aerosol formation
- Study has shown that 40% of global black carbon emissions come from biomass burning



# Fossil fuel combustion

- Burning of solid, liquid, or gas fossil fuels
  - Mainly soot, sulfate, fly ash, metals
- Many fine PMs, some coarse (fly ash)
- Diesel vehicle emit 2-20 times than gas vehicle
  - Without control 10-100 times
- NASA visualization of global aerosol/PM:  
[https://www.youtube.com/watch?v=oRsY\\_UviBPE](https://www.youtube.com/watch?v=oRsY_UviBPE)
- A snap shot image from NASA:  
<https://www.nasa.gov/image-feature/just-another-day-on-aerosol-earth>

# Particle motion

- Important movements of particles
  - *Sedimentation/Settling*
  - Brownian diffusion
  - Movement under charge

# Sedimentation

- Small particles can take a very long time to settle
  - Hence can be carried away further with wind
  - Size is very important to how far they will travel

| Particle diameter ( $\mu\text{m}$ ) | 0.02      | 0.1      | 1        | 10       | 100       | 1000      | 5000        |
|-------------------------------------|-----------|----------|----------|----------|-----------|-----------|-------------|
| Time to fall 1 km                   | 228 years | 36 years | 328 days | 3.6 days | 1.1 hours | 4 minutes | 1.8 minutes |

Table 5.4 Jacobson, 2012

# Sedimentation

- Assume spherical particles, 1 atm, 20°C
- Falling particles are mainly subjected to:

– **Gravity ( $F_g$ )**  $F_g = mg = V \rho_p g$

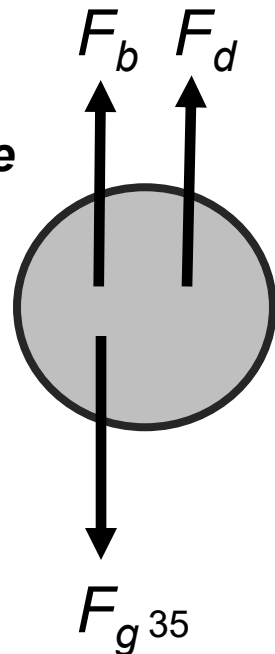
9.8 m/s<sup>2</sup>

– **Buoyancy ( $F_b$ )**  $F_b = V \rho g$  *Density of particle*

- Archimedes principle

*Density of air*  
~1.2 kg/m<sup>3</sup>

– **Drag ( $F_d$ )** *Volume of particle*



# Drag force

- Drag force ( $F_d$ )
  - Air resistant during particle falling

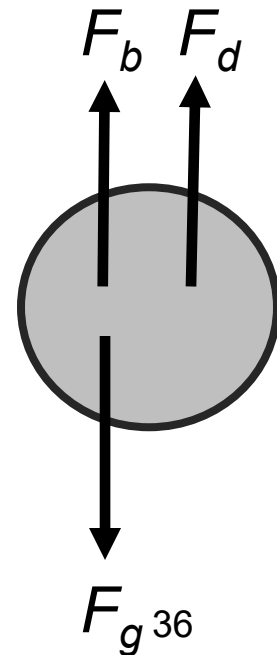
$$F_d = \frac{\pi C_D \rho D_p^2 u^2}{8C_c}$$

*Particle diameter*

*drag coefficient*

*Settling speed*

*Cunningham (slip) correction factor*



# Cunningham (slip) correction factor

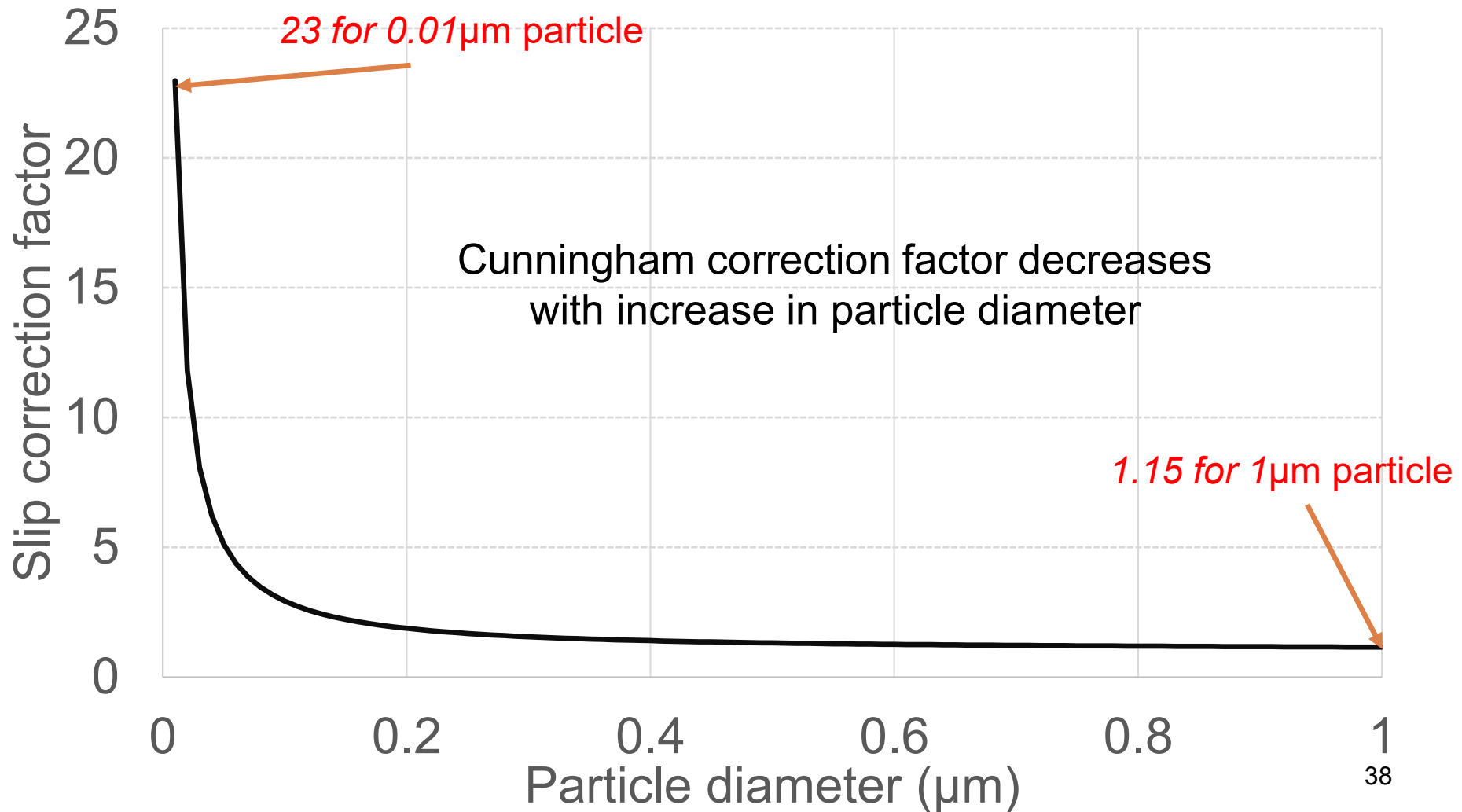
- Drag force  $\leftarrow$  air molecules hit particles
- Air molecules not attached to each other
- Small particles can slip between air molecules
  - Hence settling faster
- Empirically, Cunningham correction factor ( $C_c$ ):

$$C_c = 1 + \frac{\lambda}{D_p} \left[ 2.33 + 0.96 \exp\left(-0.5 \frac{D_p}{\lambda}\right) \right]$$



***Mean free path of air (~0.066  $\mu\text{m}$  at 1 atm and 20°C)  
average travel distance of an air molecule between successive collisions***

# Cunningham (slip) correction factor



# Drag coefficient

- Drag coefficient is a function of Reynolds number ( $Re$ )
- Reynolds number is: inertial/viscous force

$$Re = \frac{D_p \rho u}{\mu}$$

**Dynamic** viscosity of air ( $\sim 1.7 \times 10^{-5}$  kg/m/s)

- When  **$Re < 0.1$** ,  $C_D = 24/Re$ 
  - Stokes' law

# Terminal Settling Velocity

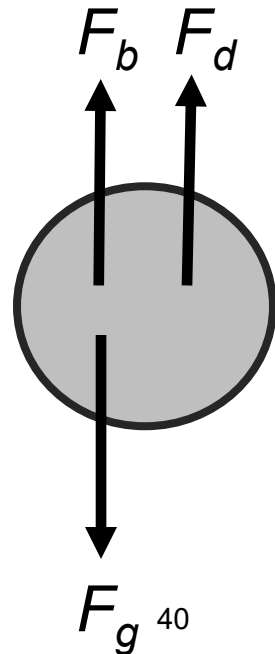
- Particle normally reaches force equilibrium quickly
- At force equilibrium:

$$F_g = F_d + F_b$$

- Solve for ***Terminal Settling Velocity***:

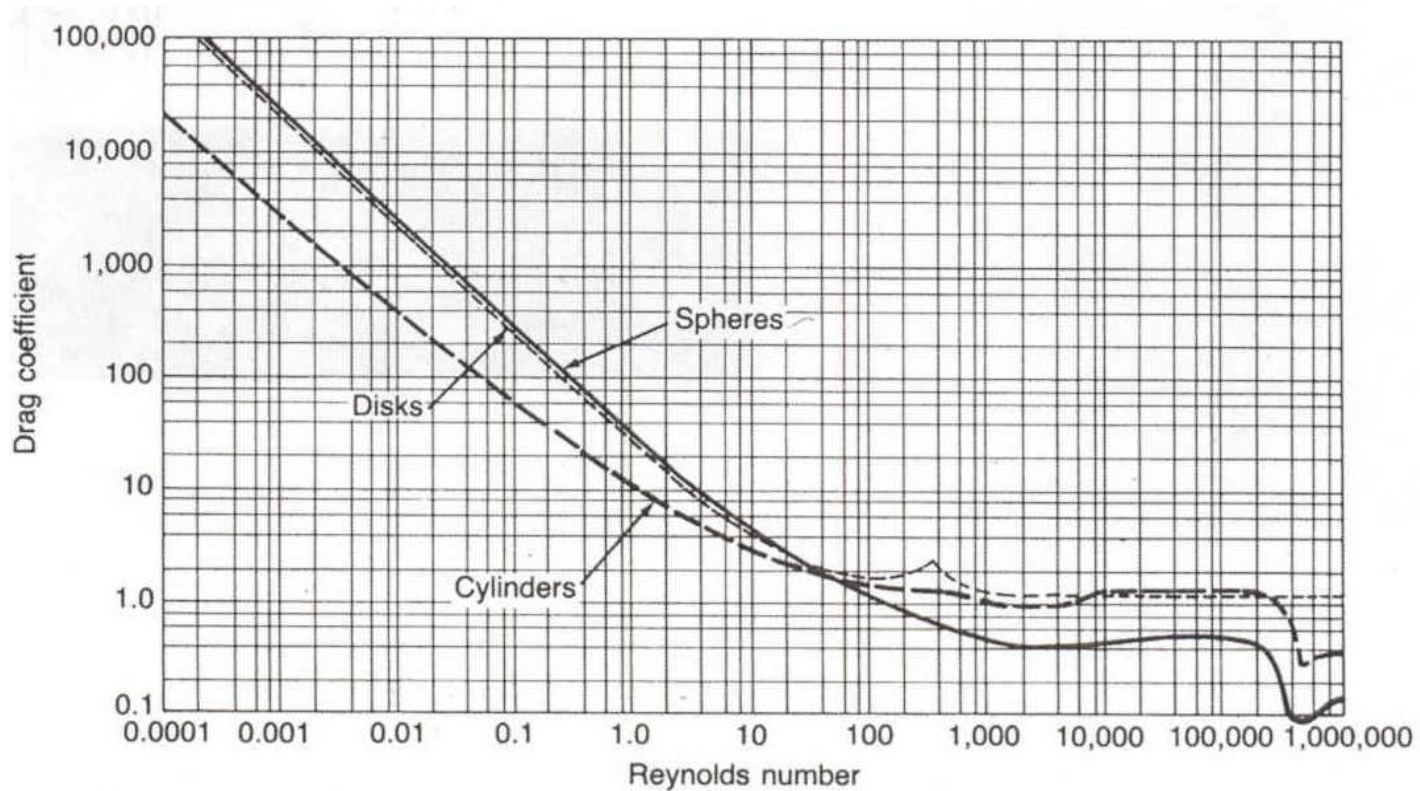
$$u_t = \frac{D_p^2 \rho_p g C_c}{18\mu}$$

Applies when  $Re < 0.1$   
Use of this equation for high  $Re$   
particles → over prediction



# At high Re number....

- $C_D$  not linear relationship with Re



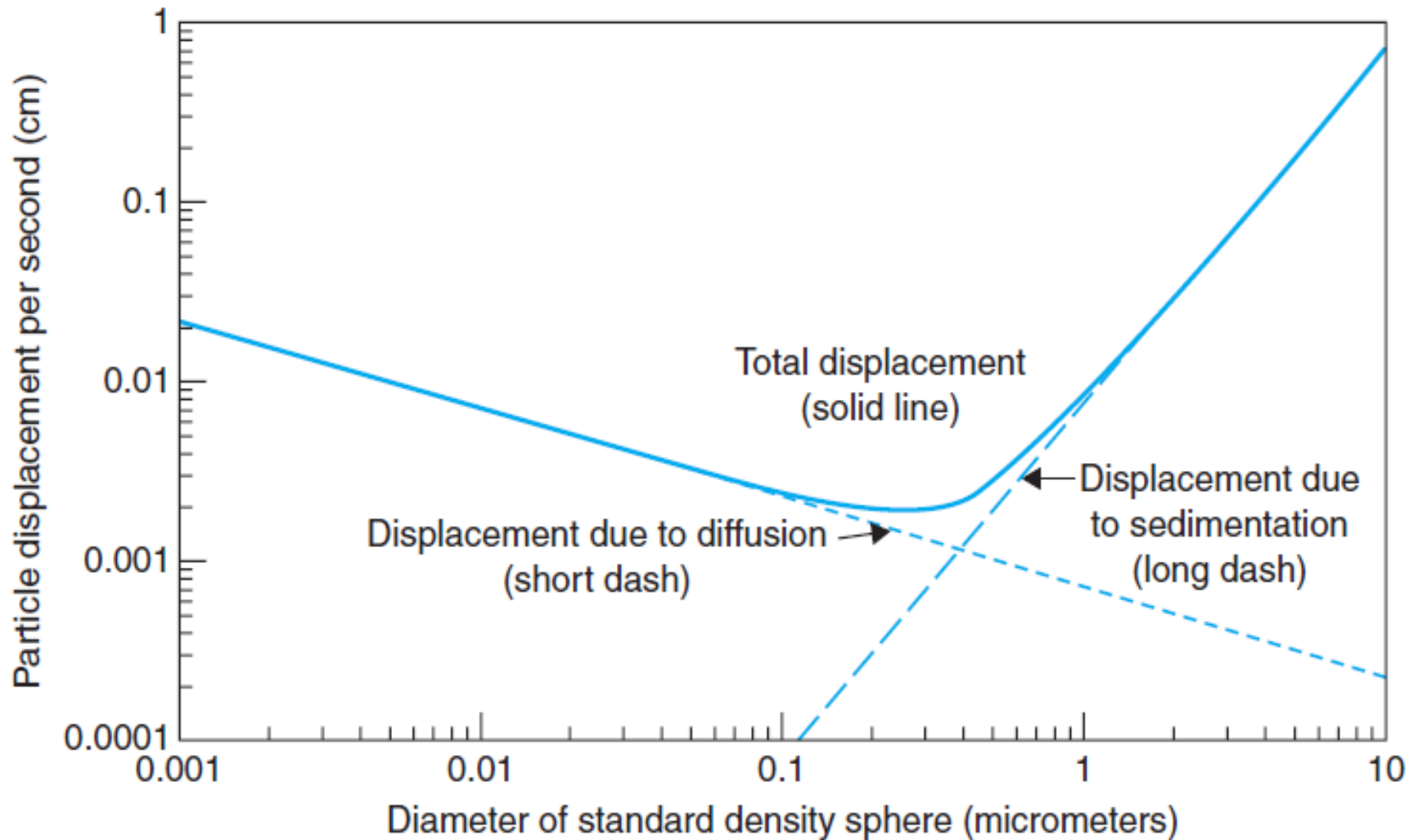
# Particle relaxation Time

| Particle Diameter ( $\mu\text{m}$ ) | Relaxation Time (s)  |
|-------------------------------------|----------------------|
| 0.01                                | $7.0 \times 10^{-9}$ |
| 0.1                                 | $9.0 \times 10^{-8}$ |
| 1.0                                 | $3.5 \times 10^{-6}$ |
| 10.0                                | $3.1 \times 10^{-4}$ |
| 100                                 | $3.1 \times 10^{-2}$ |

# Brownian diffusion

- Brownian diffusion is the random movement of particles in the air due to collision with fast-moving molecules
- Effect greater with temperature increase
- Becomes important than sedimentation for particles on the order of  $0.1\mu\text{m}$

# Brownian diffusion vs sedimentation



(Phalen, 2013)

# Movement under charge

- Particles in the environment are often charged
  - Some are highly charged, resulting forces thousands times stronger than gravity
- Due to their charge, electric forces can also act on them, impacting their movement (and deposition)
  - Usually small, but can be important in electric fields
- Certain equipment rely on the electrical properties of particles
  - Electrostatic precipitators

# Interaction of particles

- Coagulation
- Impaction/interception
- Conversion between size modes

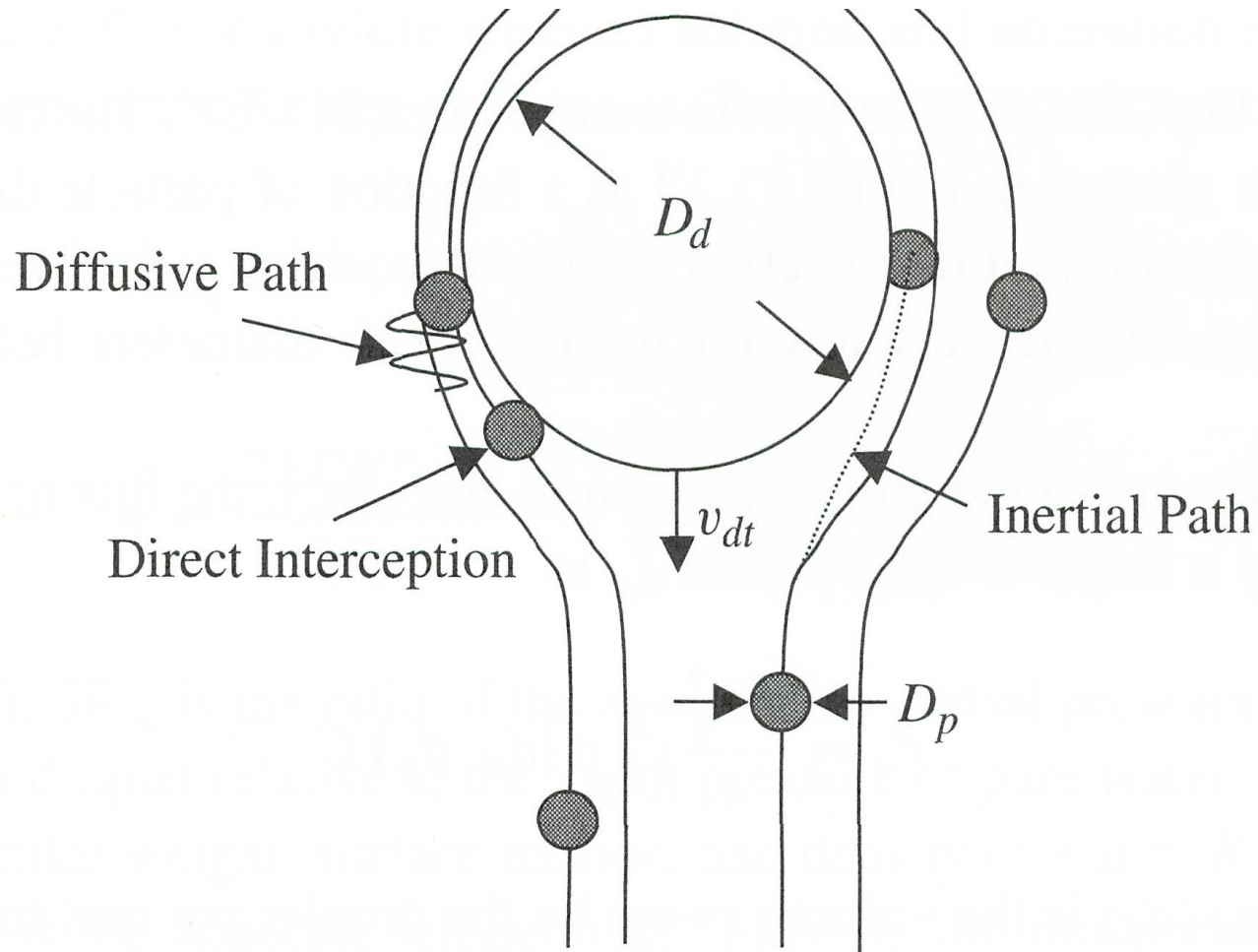
# Coagulation

- Two particles collide and stick together
- Number concentration ↓ & particle size ↑
  - Same mass concentration
- When Brownian diffusion dominates particle motion
  - Thermal coagulation
- When external forces dominates particle motion
  - Kinematic coagulation

# Impaction/interception

- Scavenging of particles by other particles or surfaces
- Impaction
  - Particles have inertia and can't follow the gas streamlines
  - Direct impact on other particles or surface
- Interception
  - Particle remains on streamline but is large enough to contact anyway

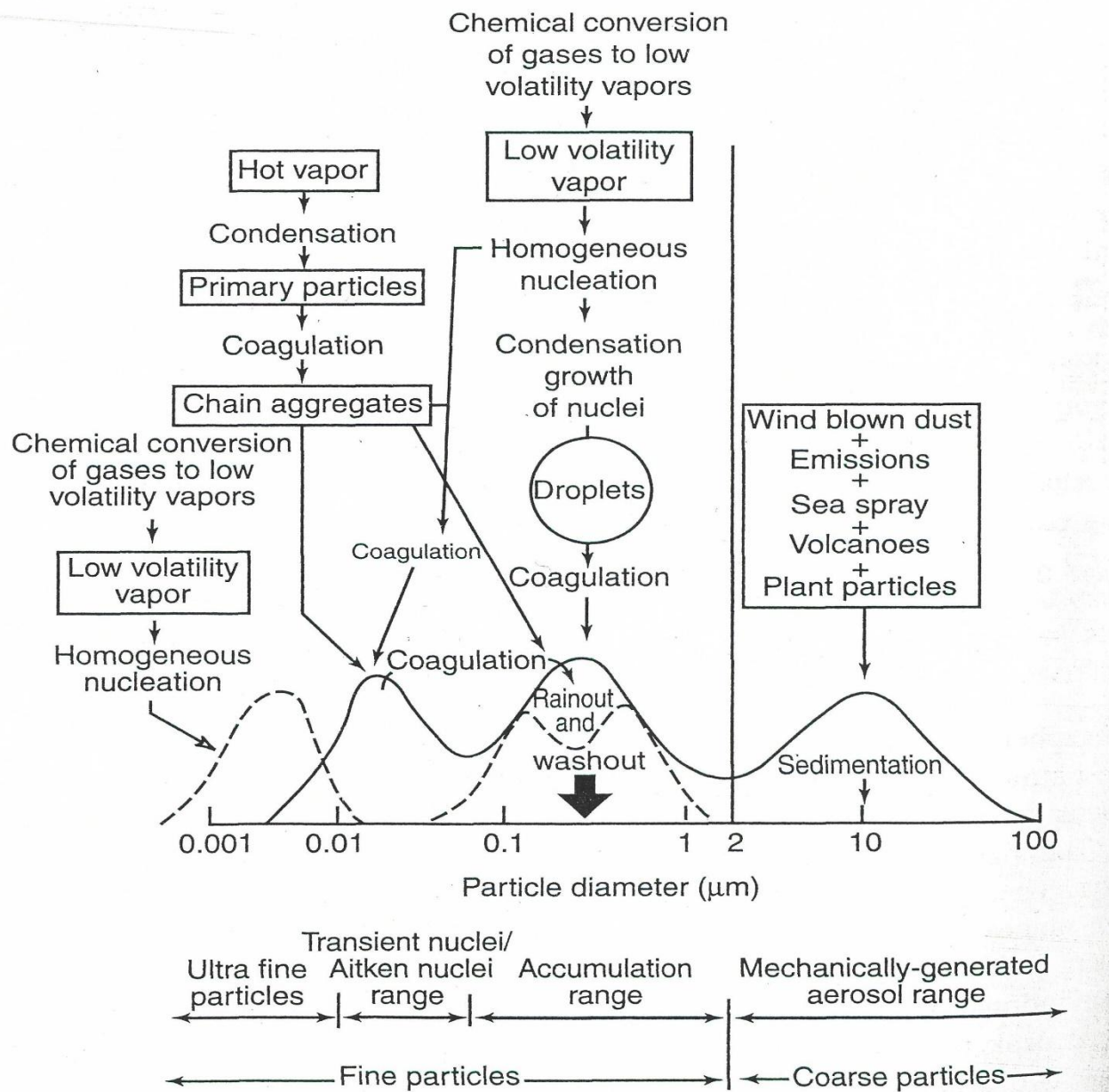
# Impaction/interception



# Conversion between size modes

- The size distribution of aerosol will change over time and space
  - Due to all of the processes mentioned previously
- Particle itself: condensation/evaporation
- Many processes affect particle size distribution

# Processes affecting particle size distribution



Source and processed of atmospheric aerosols at different size modes (Godish 2003).  
 Solid line: three mode model by Whitby & co-worker; Dash line: added ultrafine mode with sometime observed accumulation model distribution. Vertical axis: maybe number, area or mass. This figure is for illustration only